

Unit 5
Chapter 07
Chapter 7: Data visualization

7.1 GIS and maps

There is a strong relationship between maps and GIS. More specifically, maps can be used as input for a GIS. They play a key role in relation to all the functional components of a GIS.

As soon as a question contains a “where?” question, a map can often be the most suitable tool to solve the question and provide the answer. “Where do I find Enschede?” and “Where did ITC’s students come from?” are both examples.

“Where?” Of course, the answers could be in non-map form like “in the Netherlands” or “from all over the world.” These answers could be satisfying; however, they do not give the full picture.

A map would put these answers in a spatial context. It could show where in the Netherlands Enschede is to be found and where it is located with respect to Schiphol–Amsterdam airport, where most students arrive. A world map would refine the answer “from all over the world,” since it reveals that most students arrive from Africa and Asia, and only a few come from the Americas, Australia and Europe.

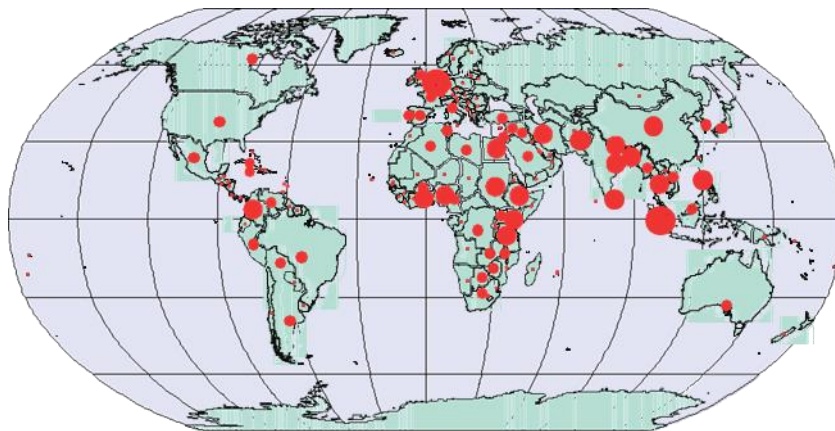


Figure: Maps and location—“Where did ITC cartography students come from?” Map scale is 1 : 200;000;000.

However, maps can do more than just providing information on location. They can also inform about the thematic attributes of the geographic objects located in the map. An example would be “What is the predominant land use in southeast Twente?” The answer could, again, just be verbal and state “Urban.” However, such an answer does not reveal patterns

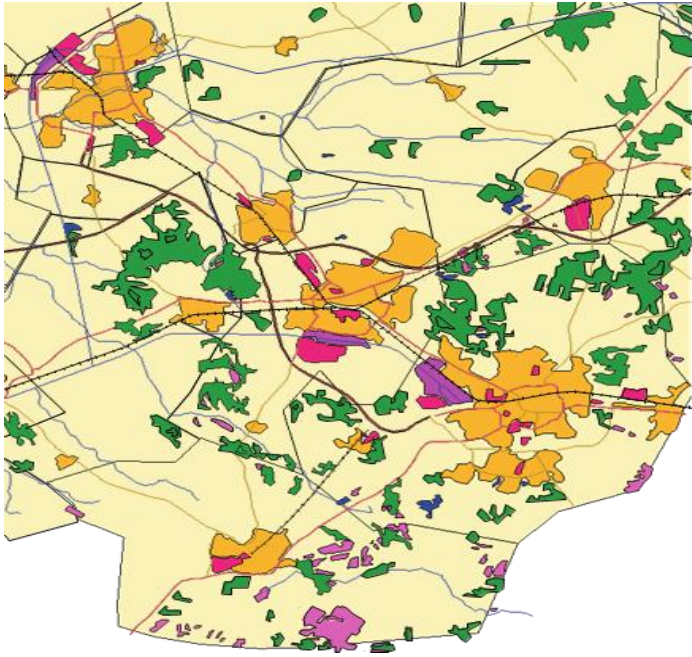


Figure: Maps and characteristics— “What is the predominant land use in southeast Twente?”

A third type of question that can be answered from maps is related to “When?” For instance, “When did the Netherlands have its longest coastline?” The answer might be “1600,” and this will probably be satisfactory to most people. However, it might be interesting to see how this changed over the years

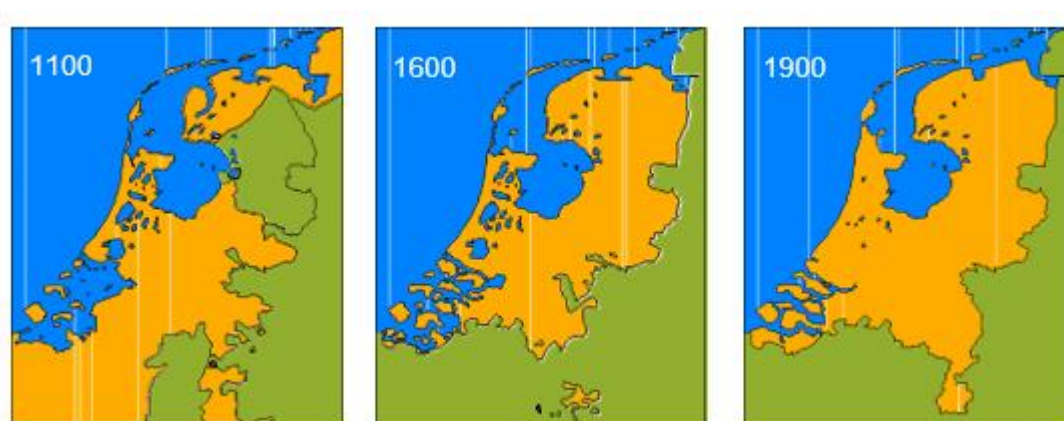


Figure: Maps and time— “When did the Netherlands have its longest coastline?”

This characteristic is well illustrated when one puts the map next to an aerial photograph of the same area. Products like these give all in Simplification and abstraction from reality formation observed by the capture devices used. Figure 7.4 shows an aerial photograph of the ITC building and a map of the same area. The photographs show all visible objects, including parked cars, and small temporary buildings. From the photograph, it becomes clear that the weather as well

as the time of the day influenced its contents: the shadow to the north of the building's obscures other information. The map on the other hand, only gives the outlines of buildings and the streets in the surroundings. It is easier to interpret because of selection/omission and classification of features.

Map scale

There is a relationship between the effectiveness of a map for a given purpose and the map's scale. The Public Works department of a city council cannot use a 1: 250;000 map for replacing broken sewer-pipes, and the map of Figure 7.1 cannot be reproduced at scale 1: 10;000. The map scale is the ratio between a distance on the map and the corresponding distance. Maps that show much detail of a small area are called large-scale maps. The map in Figure 7.4 displaying the surroundings of the ITC-building is an example. The world maps

in Figure 7.1 is a small-scale map. Scale indications on maps can be given verbally like 'one-inch-to-the-mile', or as a representative fraction like 1: 200;000;000 (1cm on the map equals 200;000;000cm (or 2;000km) in reality), or by a graphic representation like a scale bar as given in the map in Figure 7.4(b). The advantage of using scale bars in digital environments is that its length changes also when the map is zoomed in or enlarged before printing.¹ Sometimes it is necessary to convert maps from one scale to another, but this may lead to problems of (cartographic) generalization.

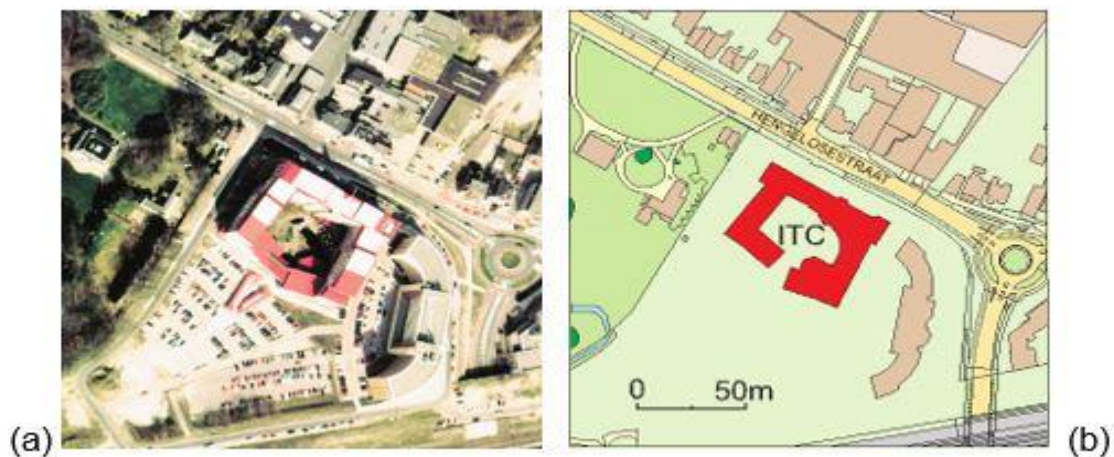


Figure: Comparing an aerial photograph (a) and a map (b)

Topographic maps

Traditionally, maps are divided into topographic maps and thematic maps. A topographic map visualizes, limited by its scale, the Earth's surface as accurately

as possible. This may include infrastructure (e.g. railroads and roads), land use (e.g. vegetation and built-up area), relief, hydrology, geographic names and a reference grid. Figure 7.5 shows a small-scale topographic map (with text omitted) of Overijssel, the Dutch province in which Enschede is located.

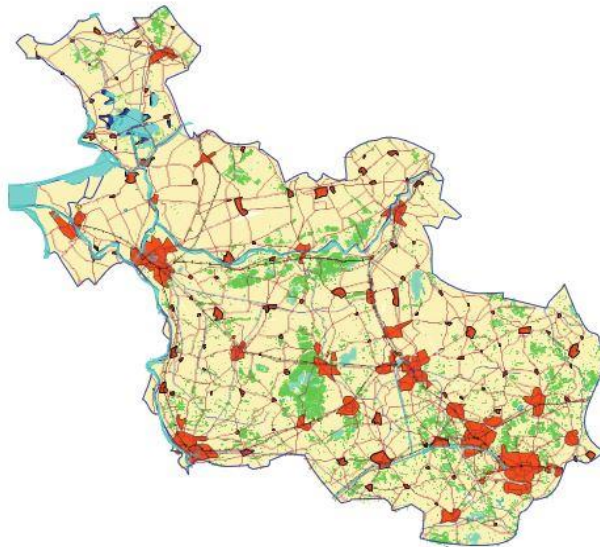


Figure 7.5: A topographic map of the province of Overijssel. Geographic names and a reference grid have been omitted for reasons of clarity

Thematic maps

Thematic maps represent the distribution of particular themes. One can distinguish between socio-economic themes and physical themes. The map in Figure 7.6(a), showing population density in Overijssel, is an example of the first and the map in Figure 7.6(b), displaying the province's drainage areas, is an example of the second. As can be noted, both thematic maps also contain information found in a topographic map, to provide a geographic reference to the theme represented. The amount of topographic information required depends on the map theme.

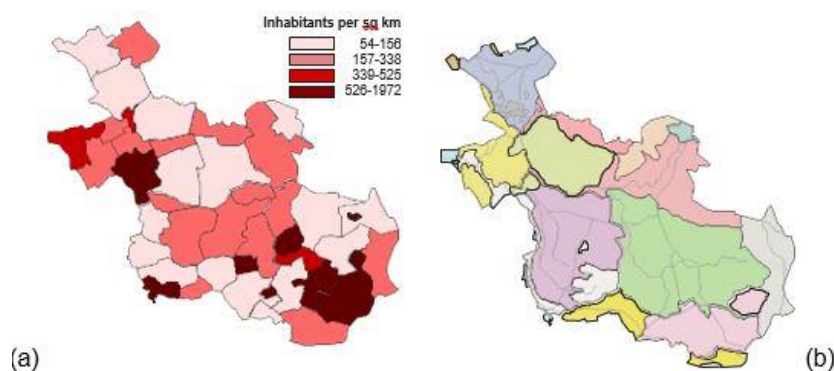


Figure 7.6: Thematic maps: (a) socio-economic thematic map, showing population density of the province of Overijssel (higher densities in darker tints); (b) physical thematic map, showing watershed areas of (b) Overijssel

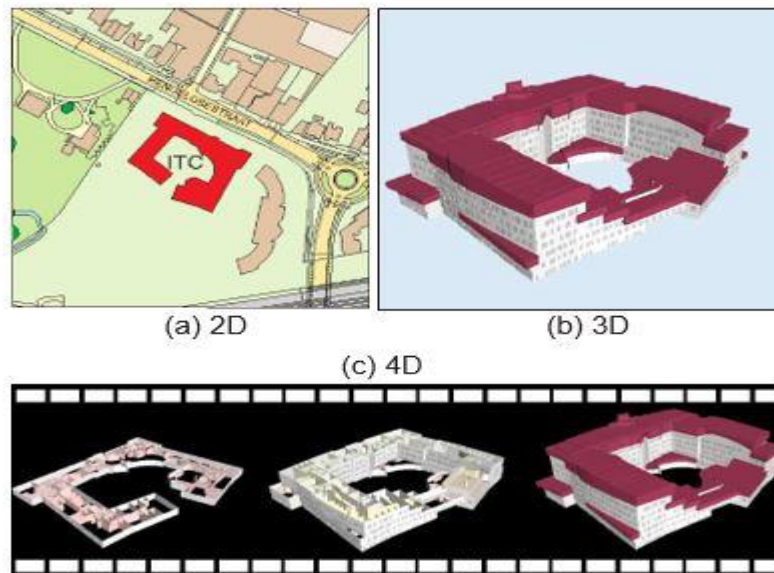


Figure 7.7: The dimensions of spatial data: (a) 2D, (b) 3D, (c) 3D with time

7.2 The visualization process

The characteristic of maps and their function in relation to the spatial data handling process was explained in the previous section. In this context the carto-graphic visualization process is considered to be the translation or conversion of spatial data from a database into graphics. These are predominantly map-like products. During the visualization process, cartographic methods and techniques are applied. These can be considered to form a kind of grammar that allows for the optimal design and production for the use of maps, depending on the application (see Figure 7.8).

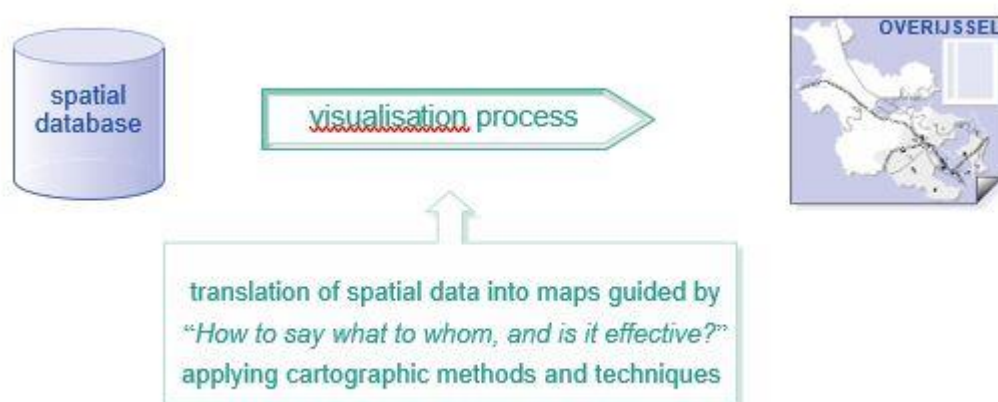


Figure 7.8: The cartographic visualization process.

The visualization process can vary greatly depending on where in the spatial data handling process it takes place and the purpose for which it is needed. Visualizations can be, and are, created during any phase of the spatial data handling process as indicated before. They can be simple or complex, while the production time can be short or long.

Visualization purpose and environment

The visualization process is guided by the question “How do I say what to whom?” “How” refers to cartographic methods and techniques. “I” represents the cartographer or map maker, “say” deals with communicating in graphics the semantics of the spatial data. “What” refers to the spatial data and its characteristics, (for instance, whether they are of a qualitative or quantitative nature). “Whom” refers to the map audience and the purpose of the map—a map for scientists requires a different approach than a map on the same topic aimed at children. This will be elaborated upon in the following sections.

In the past, the cartographer was often solely responsible for the whole map compilation process. During this process, incomplete and uncertain data often still resulted in an authoritative map. The maps created by a cartographer had to be accepted by the user. Cartography, for a long time, was very much driven by supply rather than by demand

Effectiveness

The visualization process should also be tested on its effectiveness. To the proposition “How do I say what to whom” we have to add “and is it effective?” Based on feedback from map users, or knowledge about the effectiveness of cartographic solutions, we can decide whether improvements are needed, and derive recommendations for future application of those solutions. With all the visualization options available, such as animated maps, multimedia and virtual reality, it remains necessary to test the effectiveness of cartographic methods and tools.

Cartographic rules

Cartographers have all kind of tools available to visualize the data. These tools consist of functions, rules and habits. Algorithms used to classify the data or to smooth a polyline are examples of functions. Rules tell us, for instance, to use proportional

symbols to display absolute quantities or to position an artificial light source in the northwest to create a shaded relief map. Habits or

conventions—or traditions as some would call them—tell us to colour the sea in blue, lowlands in green and mountains in brown. The efficiency of these tools will partly depend on the above-mentioned factors, and partly on what we are used to.

7.3 Visualization strategies: present or explore?

Visual communication

Traditionally the cartographer's main task was the creation of good cartographic products. This is still true today. The main function of maps is to communicate geographic information, i.e. to inform the map user about location and nature of geographic phenomena and spatial patterns. This has been the map's function throughout history. Well-trained cartographers are designing and producing maps, supported by a whole set of cartographic tools and theory as described in cartographic textbooks.

Visual thinking process

During the last decades, many others have become involved in making maps. The widespread use of GIS has increased the number of maps tremendously. Even the spreadsheet software used commonly in office today has mapping capabilities, although most users are not aware of this. Many of these maps Visual thinking process are not produced as final products, but rather as intermediaries to support the user in her/his work dealing with spatial data. The map has started to play a completely new role: it is not only a communication tool, but also has become an aid in the user's (visual) thinking process.

Visual data mining

This thinking process is accelerated by the continued developments in hard-and software. Media like DVD-ROMs and the WWW allow dynamic presentation and user interaction. These went along with changing scientific and societal needs for georeferenced data and, as such, for maps. Users now expect immediate and real-time access to the data; data that have become abundant in many sectors of the geoinformation world. This abundance of data, seen as a 'paradise' by some sectors, is a major problem in other sectors. We lack the tools for

user-friendly queries and retrieval when studying the massive amount of (spatial) data produced by sensors, which is now available via the WWW. A new branch of science is

currently evolving to deal with this problem of abundance. In the geo-disciplines, it is called visual data mining

Interaction and dynamics

These developments have given the term visualization an enhanced meaning. According to the dictionary, it means 'to make visible' or 'to represent in graphical form'. It can be argued that, in the case of spatial data, this has always been the business of cartographers. However, progress in other disciplines has linked the word to more specific ways in which modern computer technology can facilitate the process of 'making visible' in real time. Specific software toolboxes have been developed, and their functionality is based on two key words: interaction and dynamics. A separate discipline, called scientific visualization, has developed around it [37], and has also had an important impact on cartography. It offers the user the possibility of instantaneously changing the appearance of a map. Interaction with the map will stimulate the user's thinking and will add a new function to the map. As well as communication, it will prompt thinking and decision-making.

Geovisualization

Exploration is defined as 'private visual thinking' because it is often an individual playing with the spatial data to determine its significance. It is obvious that presentation fits into the traditional realm of cartography, where the cartographer works on known spatial data and creates communicative maps. Such maps are often created for multiple use. Exploration, however, often involves a discipline expert who creates maps while dealing with unknown data. These maps are generally for a single purpose, expedient in expert's attempt to solve a problem. While dealing with the data, the expert should be able to rely on cartographic expertise, provided by the software or some other means. Essentially, here the problem of translation of spatial data into cartographic symbols also needs to be solved.

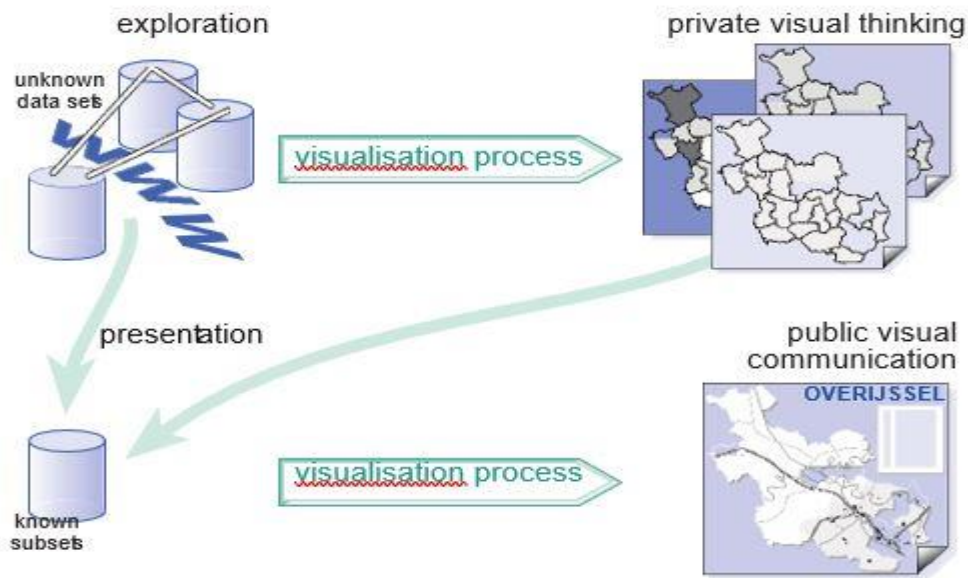


Figure 7.9: Private visual thinking and public visual communication.

Exploration means to search for spatial, temporal or spatiotemporal patterns, relationships between patterns, or trends. In case of a search for patterns, a domain expert may be interested in aspects like the distribution of a phenomenon, the occurrence of anomalies, the sequence of appearances and disappearances. A search for relationships between patterns could include changes in vegetation indices and climatic parameters, location of deprived urban areas and their distance to educational facilities. A search for trends could, for example, focus on the development in distribution and frequency of landslides. Maps not only enable these types of searches, findings may also trigger new questions, and lead to new visual exploration (or analytical) acts.

To create a map, one selects relevant geographic data and converts these into meaningful symbols for the map. Paper maps (in the past) had a dual function. They acted as a database of the objects selected from reality and communicated information about these geographic objects. The introduction of computer technology, and databases, has created a split between these two functions of the map. The database function is no longer required for the map, although each map can still function like it. The communicative function of maps has not changed.

Visual communication process

The sentence "How do I say what to whom and is it effective?" guides the cartographic visualization process, and summarizes the cartographic communication principle. Especially when dealing with maps in the realm of presentation cartography (Figure

7.9), it is important to adhere to the cartographic design rules. This is to guarantee that the resulting maps are easily understood by their users. How does this communication process work? Figure 7.10 forms an illustration. It starts with information to be mapped (the 'What' from the sentence). Before anything can be done, the cartographer should get a feel for the nature of the information, since this determines the graphical options. Cartographic information analysis provides this. Based on this knowledge, the cartographer can choose the correct symbols to represent the information in the map. S/he has a whole toolbox of visual variables available to match symbols with the nature of the data.

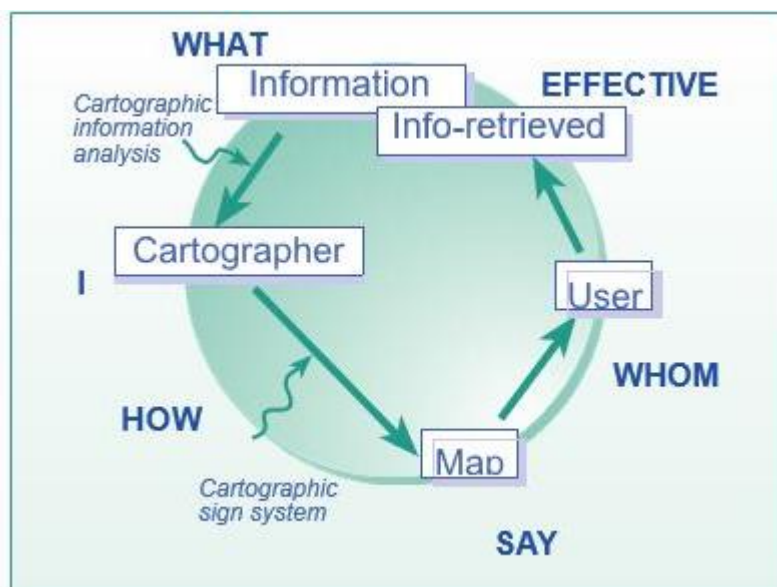


Figure 7.10: The cartographic communication process, based on "How do I say what to whom, and is it effective"

7.4 The cartographic toolbox

7.4.1 What kind of data do I have?

To derive the proper symbology for a map one has to execute a cartographic data analysis. The core of this analysis process is to access the characteristics of the data to find out how they can be visualized, so that the map user properly interprets them. The first step in the analysis process is to find a common -denominator for all the data. This common denominator will then be used as the title of the map. For instance, if all data are related to land use, collected in 2005, the title could be Land use of...2005. Secondly, the individual component(s), such as land use, and probably relief, should be analysed and their nature described. Later, these components should be visible in the map legend.

Data will be of a qualitative or quantitative nature. Qualitative data is also called nominal or categorical data. This data exists as discrete, named values without a natural order amongst the values. Examples are the different languages (e.g. English, Swahili, Dutch), the different soil types (e.g. sand, clay, peat) or the different land use categories (e.g. arable land, pasture). In the map, qualitative data are classified according to disciplinary insights such as a soil classification system represented as basic geographic units: homogeneous areas associated with a single soil type, recognized by the soil classification.

In between qualitative and quantitative data, one can distinguish ordinal data. These data are measured along a relative scale, based on hierarchies. For in-stance, one knows that one value is 'more' than another value, such as 'warm' versus 'cool'. Another example is a hierarchy of road types: 'highway', 'main road', 'secondary road' and 'track'. The different types of data are summarized in Table 7.1.

Measurement scale	Nature of data
Nominal, categorical	Data of different nature / identity of things (qualitative)
Ordinal	Data with a clear element of order, though not quantitatively determined (ordered)
Interval	Quantitative information with arbitrary zero
Ratio	Quantitative data with absolute zero

Table 7.1: Differences in the nature of data and their measurement scales

7.4.2 How can I map my data?

Basic elements of a map, irrespective of the medium on which it is displayed, are point symbols, line symbols, area symbols, and text. The appearance of point, line, and area symbols can vary depending on their nature. Most maps in this book show symbols in different size, shape and colour. Points can vary in form or colour to represent the location of shops or they can vary in size to represent aggregated values (like number of inhabitants) for an administrative area. Lines can vary in colour to distinguish between administrative boundaries and rivers or vary in shape to show the difference between railroads and roads. Areas follow the same principles: difference in colour distinguishes between different vegetation stands.

The variations in symbol appearance are only limited by the imagination they can be grouped together in a few categories. Bertin [5] distinguished six categories, which he called the visual variables, and which may be applied to point, line and area symbols

differences in:	symbols		
	point	line	area
size			
value			
grain			
colour			
orientation			
shape			

Figure 7.11: Bertin's six visual variables illustrated. Source: Plate 1 in [31]

These visual variables can be used to make one symbol different from another. In doing this, map makers in principle have free choice, provided they do not violate the rules of cartographic grammar. They do not have that choice when deciding where to locate the symbol in the map. The symbol should be located where features belong. Visual variables influence the map user's perception in different ways. What is perceived depends on the human capacity to see or perceive:

What is of equal importance (e.g. all red symbols represent danger),

Order (e.g. the population density varies from low to high—represented by light and dark colour tints, respectively),

Quantities (e.g. symbols changing in size with small symbols for small amounts), or

An instant overview of the mapped them

percepti on properti es	visual variabl es	measurement scales			
		nominal	ordinal	interval	ratio
	dimensi ons of the plane	x	x	x	x
order quantities	& Size		x	x	x
order	(grey) value grain/textur e		x x	x x	
equal importance	colour hue orientation shape	x x x			

Table: Measurement scales linked to visual variables based on perception properties

7.5 How to map ...?

The subsections in this How to map ... section deal with characteristic mapping problems. We first describe a problem and briefly discuss a solution based on cartographic rules and guidelines. The need to follow these rules and guidelines is illustrated by some maps that have been wrongly designed but are nevertheless commonly found.

7.5.1 How to map qualitative data

If, after a long fieldwork period, one has finally delineated the boundaries of a province's watersheds, one likely is interested in a map showing these areas. The geographic units in the map will have to represent the individual watersheds. In such a map, each of the watersheds should get equal attention, and none should stand out above the others.



Figure: A good ex-ample of mapping qualitative data

The application of colour would be the best solution since it has characteristics that allow one to quickly differentiate between different geographic units. However, since none of the watersheds is more important than the others, the colours used must be of equal visual weight or brightness. Figure 7.12 gives an example of a correct map. The is influenced by the number of displayed Readability geographic units. In this example, there are about 15. When this number is much higher, the map, at the scale displayed here, will become too cluttered.

Figure shows two examples of how not to create such a map. In (a), several tints of black are used—as application of the visual variable 'value'. Looking at the map may cause perceptual confusion since the map image suggests differences in importance that are not there in reality. In Figure (b), colours are used instead. However, where most watersheds are represented in pastel tints, one of them stands out by its bright colour. This gives the map an unbalanced look. The viewer's eye will be distracted by the bright colours, resulting in an unjustified weaker attention for other areas.

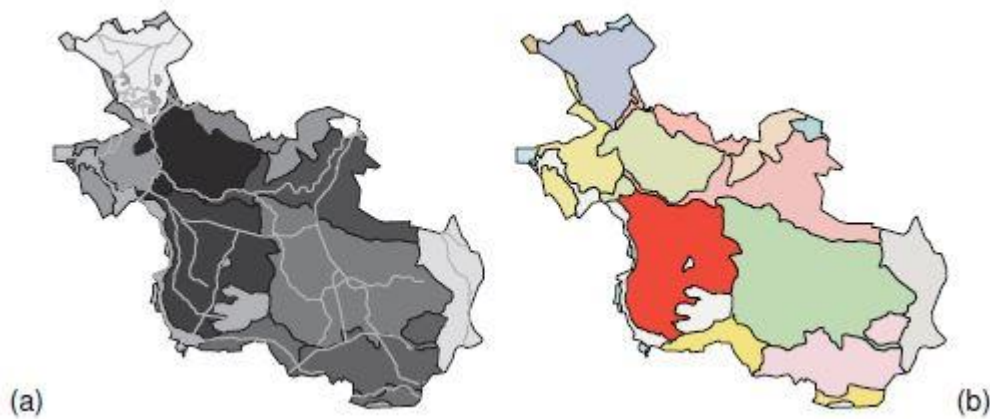


Figure: Two examples of wrongly designed qualitative maps: (a) mis-use of tints of black; (b) misuse of bright colours

7.5.2 How to map quantitative data

When, after executing a census, one would for instance like to create a map with the number of people living in each municipality, one deals with absolute quantitative data. The geographic units will logically be the municipalities. The final map should allow the user to determine the amount per municipality and offer an overview of the geographic distribution of the phenomenon. To reach this objective, the symbols used should have quantitative perception properties. Symbols varying in size fulfil this demand. Figure 7.14 shows the final map for the province of Overijssel

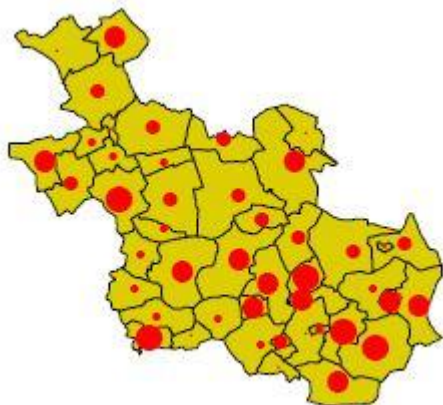


Figure: Mapping absolute quantitative data

The fact that it is easy to make errors can be seen in Figure 7.15. In 7.15(a), different tints of green (the visual variable 'value') have been used to represent absolute population numbers. The reader might get a reasonable impression of the individual amounts but not of the actual geographic distribution of the population, as the size of the geographic units will influence the perception properties too much. Imagine a small and a large unit having the same number of inhabitants. The large unit would

visually attract more attention, giving the impression there are more people than in the small unit. Another issue is that the population is not necessarily homogeneously distributed within the geographic units. Colour has also been misused in Figure 7.15(b). The applied four-colour scheme makes it impossible to infer whether red represents more populated areas than blue. It is impossible to instantaneously answer a question like "Where do most people in Overijssel live?"

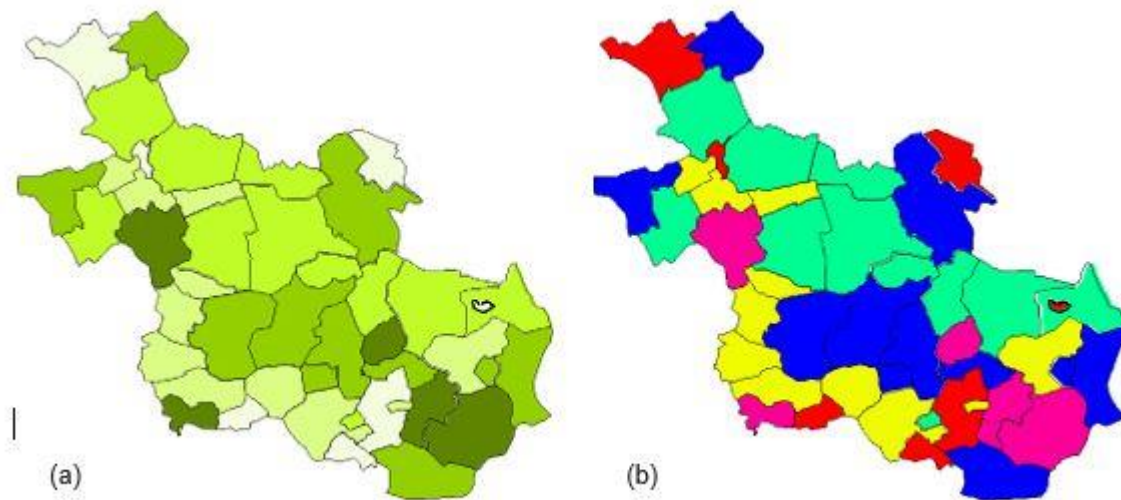


Figure: Poorly de-signed maps displaying absolute quantitative data: (a) wrong use of green tints for absolute population figures; (b) incorrect use of colour

Figure (a) shows the effect of incorrect application of the visual variable value. In this map, the value tints are out of sequence. The user must go through quite some trouble to find out where in the province the high-density areas can be found. Why should mid red represent areas with a higher population density than dark-red?

In Figure (b) colour has been used in combination with value. The first impression of the map reader would be to think the brown areas represent the areas with the highest density. A closer look at a legend would tell that this is not the case, and that those areas are represented by another colour that did not 'speak for itself'.

7.5.3 How to map the terrain elevation

Terrain elevation can be mapped using different methods. Often, one will have collected an elevation data set for individual points like peaks, or other characteristic points in the terrain. Obviously, one can map the individual points and add the height information as text. However, a contour map, in which the lines connect points of equal

elevation, is generally used. To visually improve the information content of such a map the space between the contour lines can be filled with colour and value information following a convention, e.g. green for low elevation and brown for high elevation areas. This technique is known as hypsometric or layer tinting.

Three- dimensional appearance

The shaded relief map uses the full three-dimensional information to create shading effects. This map, represented on a two-dimensional surface, can also be floated in three-dimensional space to give it a real three-dimensional appearance of a 'virtual world', as shown in Figure 7.18(d). Looking at such a presentation one can immediately imagine that it will not always be effective. Certain (low) objects in the map will easily disappear behind other (higher) objects.

Socio-economic data can also be viewed in three dimensions. This may result in dramatic images, which will be long remembered by the map user. Figure 7.19 shows the absolute population figures of Overijssel in three dimensions. Instead of a proportionally sized circle to depict the number of people living in a municipality (as we did in Figure 7.14) the proportional height of a municipality now indicates total population. The image clearly shows that Enschede (the large column in the lower right) is by far the highest populated municipality.

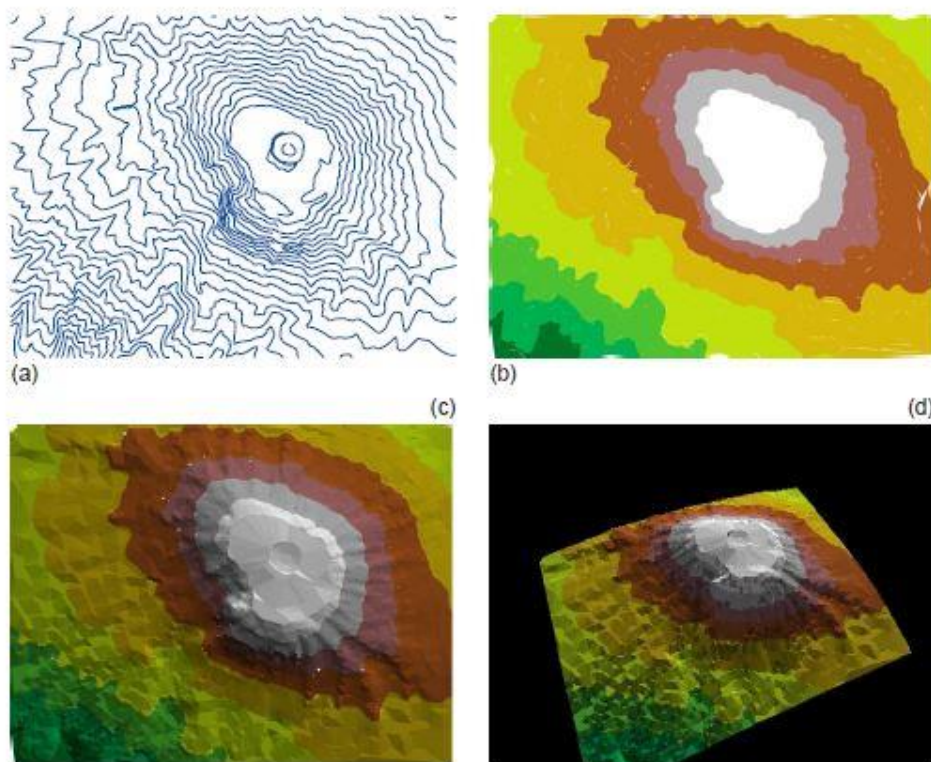


Figure : visualization of terrain elevation: (a) contour map; (b) map with layer tints; (c) shaded relief map; (d) 3D view of the terrain

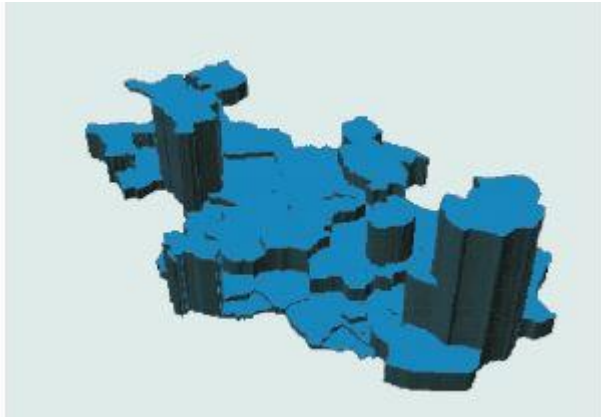


Figure: Quantitative data visualized in three dimensions

How to map time series

Advances in spatial data handling have not only made the third-dimension part of GIS routines. Nowadays, the handling of time-dependent data is also part of these routines. This has been caused by the increasing availability of data

captured at different periods in time. Next to this data abundance, the GIS community wants to analyse changes caused by real world processes. To that end, single time slice data are no longer sufficient, and the visualization of these processes cannot be supported with only static paper maps

Mapping changing phenomena

Mapping time means mapping change. This may be change in a feature's geometry, in its attributes or both. Examples of changing geometry are the evolving coast line of the Netherlands (as displayed in Figure 7.3), the location of Europe's national boundaries, or the position of weather fronts. The changes of a land parcel's owner, land use, or changes in road traffic intensity are examples of changing attributes. Urban growth is a combination of both. The urban boundaries expand and simultaneously the land use shifts from rural to urban. If maps are to represent events like these, they should be suggestive of such change.

Using symbology to represent change

This implies the use of symbols that are perceived as representing change. Examples of such symbols are arrows that have an origin and a destination. They are used to

show movement and their size can be an indication of the magnitude of change. Size changes can also be applied to other point and line symbols to show increase and decrease over time. Specific point symbols such as 'crossed swords' (battle) or 'lightning' (riots) can be found to represent dynamics in historic maps. Another alternative is the use of the visual variable value (expressed as tints). In a map showing the development of a town, dark tints represent old built-up areas, while new built-up areas are represented by light tints (see Figure 7.20(a)).

Temporal cartographic techniques

1. Single static map: Specific graphic variables and symbols are used to indicate change or represent an event. Figure 7.20(a) applies the visual variable value to represent the age of the built-up areas;
2. Series of static maps: A single map in the series represents a 'snapshot' in time. Together, the maps depict a process of change. Change is perceived by the succession of individual maps depicting the situation in successive snapshots. It could be said that the temporal sequence is represented by a spatial sequence, which the user has to follow, to perceive the temporal variation. The number of images should be limited since it is difficult for the human eye to follow long series of maps (Figure 7.20(b));
3. Animated map: Change is perceived to happen in a single image by displaying several snapshots after each other just like a video cut with successive frames. The difference with the series of maps is that the variation can be deduced from real 'change' in the image itself, not from a spatial sequence (Figure 7.20(c)).

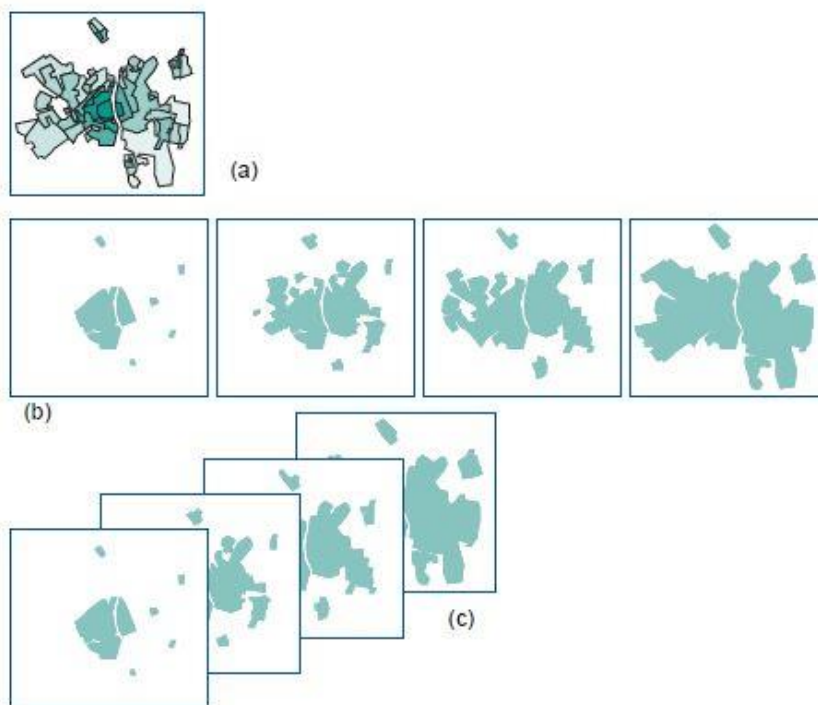


Figure: Mapping Change; example of the urban growth of the city of Maastricht, The Netherlands: (a) single map, in which tints represent age of the built-up area; (b) series of maps; (c) simulation of an animation

User interaction

For the user of a cartographic animation, it is important to have tools available that allow for interaction while viewing the animation. Seeing the animation play will often leave users with many questions about what they have seen. Just replaying the animation is not sufficient to answer questions like "What was the position of the coastline in the north during the 15th century?"

7.6 Map cosmetics

Fundamental requirements

Most maps are correct from a cartographic grammar perspective. However, many of them lack the additional information needed to be fully understood that is usually placed in the margin of printed maps. Each map should have, next to the map image, a title, informing the user about the topic visualized. A legend is necessary to understand how the topic is depicted. Additional marginal information to be found on a map is a scale indicator, a north arrow for orientation, the map datum and map

projection used, and some lineage information, (such as data sources, dates of data collection, methods used, etc.).

Further information can be added that indicates when the map was issued, and by whom (author / publisher). All this information allows the user to obtain an impression of the quality of the map and is comparable with metadata describing the contents of a database or data layer.

Space constraints

Figure 7.21 illustrates these map elements. On paper maps, these elements (if all relevant) must appear next to the map face itself. Maps presented on screen often go without marginal information, partly because of space constraints. However, on-screen maps are often interactive, and clicking on a map element may reveal additional information from the database. Legends and titles are often available on demand as well.

Text

The map in Figure 7.21 is one of the first in this chapter that has text included. Figure 7.22 is another example. Text is used to transfer information in addition to the symbols used. This can be done by the application of the visual variables to the text as well. In Figure 7.22 an example can be found. Italics—cf. the visual variable of orientation—have been used for building names to distinguish them from road names. Another common example is the use of colour to differentiate (at nominal level) between hydrographic names (in blue) and other names (in black). The text should also be placed in a proper position with respect to the object to which it refers.



Figure 7.22: Text in the map

Contrast and visual hierarchy

Maps constructed via the basic cartographic guidelines are not necessarily visually appealing maps. Although well-constructed, they might still look sterile. The design aspect of creating appealing maps also has to be included in the visualization process. 'Appealing' does not only mean having nice colours.



Figure 7.23: Visual hierarchy and the location of the ITC building: (a) hierarchy not applied; (b) hierarchy applied

One of the keywords here is contrast. Contrast will increase the communicative role of the map since it creates a hierarchy in the map contents, assuming that not all information has equal importance. This design trick is known as visual hierarchy or the figure-ground concept. The need for visual hierarchy in a map is best understood when looking at the map in Figure 7.23(a), which just shows lines. The map of the ITC building and surroundings in part (b) is an example of a map that has visual hierarchy applied. The first object to be noted will be the ITC building (the darkest patches in the map) followed by other buildings, with the road on a lower level and the parcels at the lowest level.

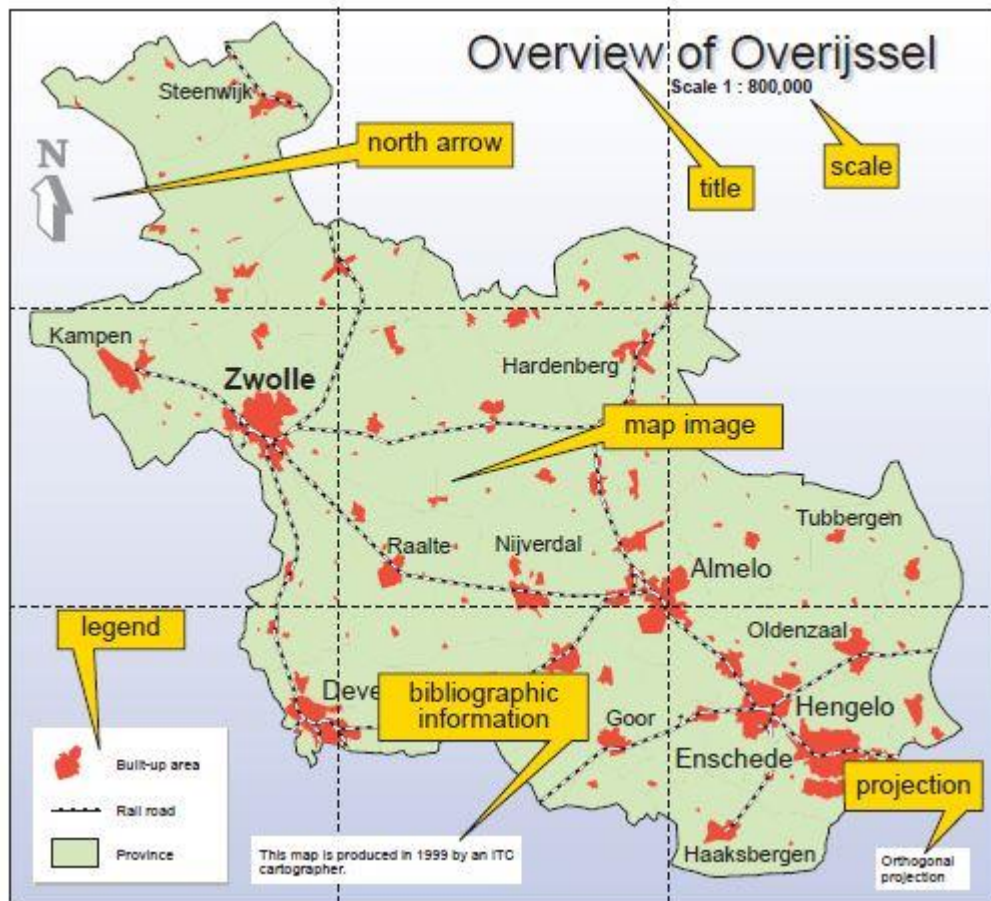


Figure 7.21: The paper map and its (marginal) information. Source: Figure 5–10 in [30].

7.7 Map dissemination

On-screen maps

Currently, most maps are presented on screen, for a quick view, for an internal presentation or for presentation on the WWW. Compared to maps on paper, on-screen maps must be smaller, and therefore their contents should be carefully selected. This might seem a disadvantage, but presenting maps on-screen offers very interesting alternatives. In one of the previous paragraphs, we discussed that the legend only needs to be a mouse click away. A mouse click could also open the link to a database, and reveal much more information than a paper map could ever offer. Links to other than tabular or map data could also be made available.

Multimedia maps

Maps and multimedia (photography, sound, video, animation) can be integrated. Some of today's electronic atlases, such as the Encarta World Atlas are good examples

of how multimedia elements can be integrated with the map. Pointing to a country on a world map starts the national anthem of the country or shows its flag. It can be used to explore a country's language; moving the mouse would start a short sentence in the region's dialects. The World Wide Web is nowadays a common medium used to present and disseminate spatial data. Here, maps can play their traditional role, for instance to show the location of objects, or provide insight into spatial patterns, but because of the nature of the internet, the map can also function as an interface to additional information. Geographic locations on the map can be linked to Maps as visual interfaces photographs, text, sound or other maps, perhaps even functions such as on-line booking services. Maps can also be used as 'previews' of spatial data products to be acquired through a spatial data clearinghouse that is part of a Spatial Data Infrastructure.

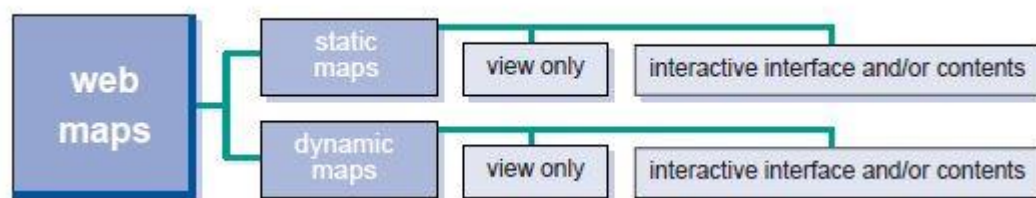


Figure 7.24: Classification of maps

Static maps

An important distinction is the one between static and dynamic maps. Many static maps on the web are view-only. Organizations, such as map libraries or tourist information providers, often make their maps available in this way. This form of presentation can be very useful, for instance, to make historical maps more widely accessible. Static, view-only maps can also serve to give web surfers a preview of the products that are available from organizations, such as National Mapping Agencies. When static maps offer more than view-only functionality, they may present an interactive view to the user by offering zooming, panning, or hyperlinking to other information. The much-used 'clickable map' is an example of the latter and is useful to serve as an interface to spatial data. Clicking on geographic objects may lead the user to quantitative data, photographs, sound or video or other information sources on the Web. The user may also interactively determine the contents of the map, by choosing data layers, and even the visualization parameters, by choosing symbology and colours

Dynamic maps

Dynamic maps are about change in one or more of the spatial data components. On the WWW, several options to play animations are available. The so-called animated-GIF can be seen as a view-only version of a dynamic map. A sequence of bitmaps, each

representing a frame of an animation, are positioned one after another, and the WWW-browser will continuously repeat the animation. This can be used, for example, to show the change of weather over the last day.

Interactive maps

Slightly more interactive versions of this type of map are those to be played by media players, for instance those in QuickTime format, or as a Flash movie. Plug-ins to the web browser define the interaction options, which are often limited to simple pause, backward and forward play. Such animations do not use any specific WWW-environment parameters and have equal functionality in the desktop-environment. The WWW also allows for the fully interactive presentation of 3D models. The Virtual Reality Mark-up Language (VRML), for instance, can be used for this purpose. It stores a true 3D model of the objects, not just a series of 3D views.

Questions:

1. What is topographic map?
2. What is thematic map?
3. What is visualization process?
4. What are the factors which influenced visualisation process?
5. What do you mean by democratization of cartography?
6. Which are the six visual variables which to distinguish cartographic symbol from each other?
7. How to map quality data?
8. How quantity data can be mapped?
9. In what way mapping time series can be done?
10. What are the fundamental requirements which comes under map cosmetics?

Multiple Choice Questions

1) Symbol on map that represent human and physical features on surface of Earth is classified as _____.

- A. legend**
- B. grid
- C. key
- D. both a and C

- 2) Compass is usually used to find the _____.
- A. movement of Moon
 - B. movement of Earth
 - C. direction**
 - D. distance
- 3) Scale of map such as 1 : 10,000 is an example of scale _____.
- A. by words
 - B. by the ratio**
 - C. by the line
 - D. by arrows
- 4) Considering grid lines of maps, horizontal lines are classified as _____.
- A. northings**
 - B. southings
 - C. eastings
 - D. westings
- 5) Symbols that are used to represent hospitals, schools and churches are classified as _____.
- A. line symbols
 - B. point symbols**
 - C. height symbols
 - D. area symbols
- 6) Numbers of eastings of map increases towards _____.
- A. south
 - B. north
 - C. west
 - D. east**
- 7) Series of horizontal and vertical lines on map is classified as _____.
- A. key lines
 - B. grid lines**
 - C. scale lines
 - D. legend lines
- 8) Basic elements of maps include _____.
- A. title and scale
 - B. grid lines and north arrow

C. legend and names of features

D. All of above

9) Four basic points on compass are classified as _____.

A. northings

B. cardinal points

C. legend points

D. eastings

10) Four cardinal points can further be divided into _____.

A. eight cardinal points

B. ten cardinal points

C. twelve cardinal points

D. six cardinal points

11) _____ are those to be played by media players, for instance those in QuickTime format, or as a Flash movie.

A. Static Maps

B. Interactive Maps

C. 2D Maps

D. 3D Maps

12) _____ are about change in one or more of the spatial data components.

A. Static Maps

B. Interactive Maps

C. Dynamic Maps

D. 3D Maps

13) _____ on the web are view-only. Organizations, such as map libraries or tourist information providers, often make their maps available in this way.

A. Static Maps

B. Interactive Maps

C. Dynamic Maps

D. 3D Maps

14) In _____ photography, sound, video, animation can be integrated. Some of today's electronic atlases, such as the Encarta World Atlas are good examples of how multimedia elements can be integrated with the map.

A. Multimedia Maps

B. Interactive Maps

C. Dynamic Maps

D. 3D Maps

15) In _____, change is perceived to happen in a single image by displaying several snapshots after each other just like a video cut with successive frames.

A. Animated Maps

B. Interactive Maps

C. Dynamic Maps

D. 3D Maps